

Data Analytics and Predictive Modelling enabling Dynamic Pricing on Data of two EV Charging Sites

Final Report of the Project Assignment - Team 6



Authors:

Linus Baudenbacher (7431174),

Nils Bringewatt (7443820),

Daniel Schumacher (7409287),

Robin Ziemen (7380337)

Lecturer: Univ.-Prof. Dr. Wolfgang Ketter

Co-Lecturer: Julius Kuhmann

Department of Information Systems for Sustainable Society

Faculty of Management, Economics and Social Sciences

University of Cologne

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Linus Baudenbacher (7431174),

Nils Bringewatt (7443820),

Daniel Schumacher (7409287),

Robin Ziemer (7380337)

Köln, den 08.02.2026

Executive Summary

The ongoing transition to electric vehicles (EVs) for domestic transportation is increasing the need for efficient and well-planned charging infrastructure. This data science project analyzes the charging session data of two distinct EV charging sites to help the operator make informed business-related decisions.

The business case focuses on enabling the implementation of dynamic pricing using discount rates, dependent on the prediction of values for the hourly site utilization. The primary data mining objective is to evaluate charging site performance, identify site characteristics, characterize the customer base, and predict utilization rates using machine learning techniques.

Data preparation steps such as cleaning, duplicate removal and erroneous value handling are carried out to ensure data quality. Key performance indicators (KPIs), including rates for utilization, energy delivery, and user registration are defined to evaluate site performance. Site characteristics are then analyzed to differentiate the company site from the university site. A clustering algorithm is applied to characterize the customer base. To enable the business case, a ridge regression model is developed to predict hourly utilization rates for the university site. The model is used to predict values on unseen data. Exemplary discount rates are computed and visualized.

In conclusion, this project illustrates how data-driven analysis can improve informed decision-making regarding the operation of EV charging sites. The model provides sufficient predictive power while being limited to the assumption, that demand patterns have recovered after the pandemic. Future modelling could include the usage of current data to avoid limitations and increase prediction power.

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1 Problem Description

Domestic transportation is one of the biggest contributors to greenhouse gas emissions in the European Union. A key element to reduce these emissions is the transition from internal combustion engines (ICE) to electric vehicles (EV). However, this transition requires a reliable, well-managed infrastructure of charging sites. Operators need to be provided with information on site utilization and user behavior to efficiently use limited capacity while optimizing revenue and ensuring adoption. In this data science project, we analyze the session data of two charging sites, aiming to gain relevant information and enabling the site operator to make informed business-related decisions.

1.1 Business Case

We aim to provide the operator with valuable insights into site utilization and user behavior. Furthermore, we want to develop a prediction model that forecasts hourly rates for site utilization to support a dynamic pricing approach. When utilization is expected to be low, discounts can be offered to registered users to enhance user engagement while smoothing demand and maximizing energy throughput and revenue.

1.2 Data Mining Goal

We aim to clean and process the raw data, derive time-dependent key performance indicators (KPIs) that characterize utilization rates, energy throughput and user registrations, explore data patterns to differentiate the public from the private site, separate the user base into clusters based on charging behavior, and develop the prediction model.

2 Data Description

The dataset contains EV charging sessions from two distinct charging sites, one available for the public at a university, and one private site at a company. Both sites are operated by the same operator. Each record includes identifiers for the session, site, charging station, and the user, if registered. Timestamps capture the session start, charging completion, and disconnection. The amount of energy delivered is recorded in kWh. Registered users may submit additional information in advance, specifying vehicle efficiency, requested energy and miles, expected session duration, and departure time. A full column overview is provided in Figure 1.

3 Data Preparation

Categorical identifiers were cast to lower case strings whereas numerical values were cast to floats and integers. Timestamps were transformed to the local timezone “America/Los Angeles” making the column “timezone” obsolete. The yearly identifier in the first column was removed since “sessionID” will be used for charging sessions. Missing “doneChargingTime” values were imputed with “disconnectTime” to avoid discarding 6% of entries of the dataset. For missing “userID” values, placeholder values were used. Checks on “sessionID” identified 1,413 duplicates. Only the latest record per session was retained. Checks on “id” and the physical key (“siteID”, “spaceID”, “stationID”, “connectionTime”) validated the removal of the duplicates. Temporal consistency checks resulted in the removal of 29 sessions, where “doneChargingTime” was logged before “connectionTime”. Furthermore, 4,692 sessions with “doneChargingTime” exceeding “disconnectTime” were clipped to “disconnectTime”. User inputs were provided as a JSON-like string. They were extracted, prepared in a similar manner and then merged back with the session data. Only the latest user input per session was kept before merging. Calendar features were extracted from “connectionTime”, duration variables were computed from timestamp differences and a binary flag was added for registered users. Outliers were not removed as they resemble variance in real world scenarios. A description of the resulting dataframe is presented in Table 1.

4 Data Analytics

4.1 Key Performance Indicators

KPI 1: Utilization Rate

The utilization rate is the indicator for resource efficiency. Keeping the rate at around 80% is preferable, as this allows new customers to be acquired while using limited capacity efficiently. Values were computed as seen in Equation 1. The majority of utilization happens during the day (Figure 2). Weekdays see much higher rates in utilization compared to weekends (Figure 3). This finding is consistent with the working hours of a company and a university. Daily utilization peaks in late 2018, keeping a consistent level until spring 2020, where we see a rapid decline caused by the corona pandemic. After the pandemic, we see a slow growth until the end of the observation window, indicating recovery (Figure 4).

KPI 2: Energy delivered per hour

This KPI reflects real charging demand rather than just site capacity. Energy throughput is also directly transformable into revenue, making it a valuable indicator for economical success. Values were computed as seen in Equation 2. Comparing values for each year, it shows that the year 2019 was the most successful, which is likely due to the impact of the pandemic (Figure 5). Comparing values for each site, it was discovered that energy throughput on site 2 is generally higher during peak hours but energy throughput on site 1 spans over longer time periods (Figure 6).

KPI 3: Rate of registered user sessions

The rate of registered user sessions serves as an indicator for user adoption, since a registered user is more likely to return. A high registration rate also enables the operator to carry out promotions to a bigger user base. Around three quarters of sessions are carried out by registered users (Figure 7). Plotting data over time, it is revealed that, after a small adoption phase in 2018, almost all users are registered with the application (Figure 8), indicating usage motivation and high potential for our business case.

4.2 Site Characteristics

The dataset contains records from a private company site and a public university site. While the session data is labeled, the site identities are not specified. The strongest evidence for *Site 1* being private is the timing of charging sessions. There is a sharp peak on weekdays between 6:00 AM and 8:00 AM, with next to no activity on weekends (Figures 9, 11). This suggests employees plugging in at the start of a workday. In contrast, *Site 2* has a less extreme peak between 8:00 AM and 10:00 AM, more pronounced evening use, and active charging on weekends (Figures 10, 11), which suggests public availability. High energy use combined with long session durations at *Site 1* further support this conclusion by indicating, that cars are parked for full work shifts rather than short, public visits (Figures 12, 13). Furthermore, *Site 1* has a much higher rate of registered and returning users (Figures 14, 15), pointing to a stable user base of employees. Based on these patterns, we conclude that *Site 1* is the private company site and *Site 2* is the public university site.

4.3 Cluster Analysis

Our clustering approach seeks to identify behavioral patterns in the data to give the operator a better understanding of the customer base. We analyzed the correlation matrix (Figure 16) of user-level features and selected variables across four dimensions: *Energy & Demand*, *Frequency & Duration*, *Temporal Behavior* and *Charging Efficiency*. We chose a Gaussian Mixture Model (GMM) for its ability to handle overlapping, elliptical clusters through soft assignments, which is a better fit for user behavior than the hard boundaries of KMeans. The optimal number of clusters ($k = 4$) was determined using BIC scores (Figure 17), which show diminishing returns for higher values for k . The Silhouette Score (Figure 18) validated the number of clusters. Each resulting cluster (Figures 19, 20, 21) exhibits a distinct behavioral signature:

Cluster 1 – Brief Infrequent Users have the lowest frequency, shortest sessions, and zero idle time. They also request far more energy than what is actually delivered.

Cluster 2 – Employees have a high session frequency and sessions with the earliest start hour and highest idle time, treating the station as a charging and parking facility.

Cluster 3 – Frequent Top-Ups is the most active user group with the highest session frequency. They have the lowest amount of energy delivered, suggesting a "top-up".

Cluster 4 – Occasional Whales receive the highest energy amount, generating a lot of revenue at once. They have short, moderately frequent sessions and low idle time.

4.4 Utilization Prediction

In this section, the development of a ridge regression model predicting hourly utilization rates at the public site will be described. A decision tree model (A.3.7) and a neural network model (A.3.8) were also tried but not used due to inferior performance.

Feature Engineering and Modelling Approach:

A flag feature "during_covid" was created to indicate the decline in site utilization at the start of the pandemic (Figure 22). Cyclic features were added for the hour, day of the week, and month to make the model aware of the cyclical relationship of time-related data. Another flag feature was added to make the model aware of public holidays. LASSO Feature selection (Figure 23) was used to identify the most useful features, resulting in ten features being selected for model training (Table 2). To imple-

ment cross-validation, the dataset was split into training (50%), validation (20%) and testing data (30%). We trained linear, lasso and ridge regression models with varying polynomial degrees in a range from 1 to 8 and values for alpha (regularization parameter) in a range from 0.01 to 10. We then compared their predictive power using R^2 as a metric on the holdout set. This methodology allowed us to select optimal hyperparameters, resulting in a ridge regression model with a polynomial degree of 5 and alpha at 10 (Table 3). The model achieved sufficient performance on the test set with an R^2 score of 0.80 a RMSE of 0.09 and a MAE of 0.06 (Table 4) while suggesting a good fit by giving comparable results for the holdout and test data (Table 5).

Model Application:

The model was used to predict values for the first week of July 2023, assuming no residual covid impact. The model made reasonable predictions for daytimes while also detecting holidays and weekends. However, it predicts utilization during night hours, suggesting predictive inaccuracy as there was almost none in the training data (Figure 24). Peak utilization on a business day is predicted to be around 50% (Figure 25). Assuming that the optimal load for a site is at 80%, we can give out discounts to registered users as an incentive, when the load is predicted to be lower. The discount given out is 1% for every 10% that we are off the optimal load. The discount rate counteracts the utilization rate (Figure 26), which can result in an increase of utilization in low usage periods, smoothing demand while motivating adoption.

5 Conclusion

Key performance indicators display important information regarding the efficiency, economical success and adoption of a charging site. Separating the customer base into clusters helps to find patterns in utilization behavior. The regression model predicts sufficient values, but relies heavily on data recorded before the pandemic. It is advised to only use the model if utilization rates have recovered. The predictions can be used to offer discounts to registered users, motivating usage during periods of low demand while smoothing demand during peak hours. All business and data mining goals of this project were successfully accomplished. Future modelling could include the usage of current data to avoid limitations and increase prediction power.

A Appendix

A.1 Figures referenced in "Data Description"

field	type	description
id	string	Unique identifier of the session record
connectionTime	datetime ^a	Time when the EV plugged in.
disconnectTime	datetime ^a	Time when the EV unplugged.
doneChargingTime	datetime ^a	Time of the last non-zero current draw recorded.
kWhDelivered	float	Amount of energy delivered during the session.
sessionID	string	Unique identifier for the session.
siteID	string	Unique identifier for the site.
spaceID	string	Unique identifier of the parking space.
stationID	string	Unique identifier of the charging point.
timezone	string	Timezone of the site. Based on pytz format.
userID	string	Unique identifier of the user, if provided.
userInputs	list	Inputs provided by the user. Since inputs can be changed over time, there can be multiple user input objects in the list.
WhPerMile ^b	float	Efficiency of the EV in Wh per mile.
kWhRequested ^b	float	Energy requested by the user in kWh.
milesRequested ^b	float	Number of miles requested by the user.
minutesAvailable ^b	float	Length of the session as estimated by the user.
modifiedAt ^b	datetime ^a	Time this user input was provided.
paymentRequired ^b	bool	If the user was required to pay for this session.
requestedDeparture ^b	datetime ^a	User estimated departure time.

^a All datetimes are in UTC (GMT); see timezone field for the correct timezone of the site.

^b Fields are optional, if user participates in app-based charging.

Figure 1: Description of the dataset provided in the problem statement

A.2 Tables referenced in "Data Preparation"

	Dimension	dtype	Non-Null Count
0	id	object	
1	connectionTime	datetime64[ns, America/Los_Angeles]	65006 non-null
2	disconnectTime	datetime64[ns, America/Los_Angeles]	65006 non-null
3	doneChargingTime	datetime64[ns, America/Los_Angeles]	65006 non-null
4	kWhDelivered	float64	65006 non-null
5	sessionID	object	65006 non-null
6	siteID	object	65006 non-null
7	spaceID	object	65006 non-null
8	stationID	object	65006 non-null
9	userID	object	65006 non-null
10	year	int32	65006 non-null
11	month	int32	65006 non-null
12	day	int32	65006 non-null
13	hour	int32	65006 non-null
14	dayofweek	int32	65006 non-null
15	is_weekend	int64	65006 non-null
16	season	object	65006 non-null
17	session_duration	float64	65006 non-null
18	charging_duration	float64	65006 non-null
19	isRegisteredUser	bool	65006 non-null
20	WhPerMile	float64	47815 non-null
21	kWhRequested	float64	47815 non-null
22	milesRequested	float64	47815 non-null
23	minutesAvailable	float64	47815 non-null
24	modifiedAt	datetime64[ns, America/Los_Angeles]	47815 non-null
25	paymentRequired	object	47815 non-null
26	requestedDeparture	datetime64[ns, America/Los_Angeles]	47815 non-null

Table 1: Finalised dataset after prepaton and cleaning

A.3 Figures, Tables and Equations referenced in "Data Analytics"

A.3.1 KPI 1: Utilization Rate

$$U_T = \left(\sum_{s=1}^n \frac{t_{sT}}{t} \right) \cdot \frac{1}{n} \quad (1)$$

T = Timeframe in observation window

U_T = Utilization rate in t

t = Duration of t

t_{sT} = Duration, station s was occupied in T

s = Station s

n = Total number of stations

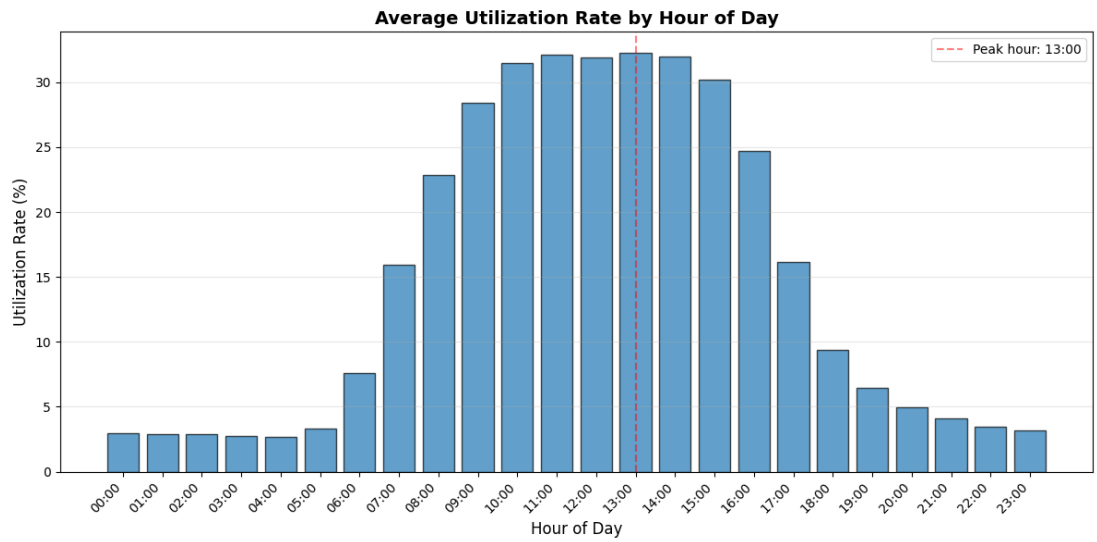


Figure 2: Average Utilization Rate by Hour of Day

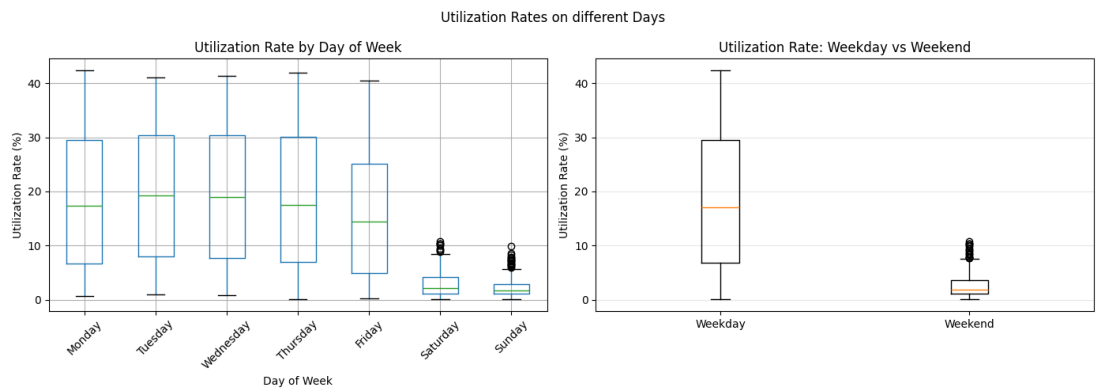


Figure 3: Average Utilization Rate by Day of Week

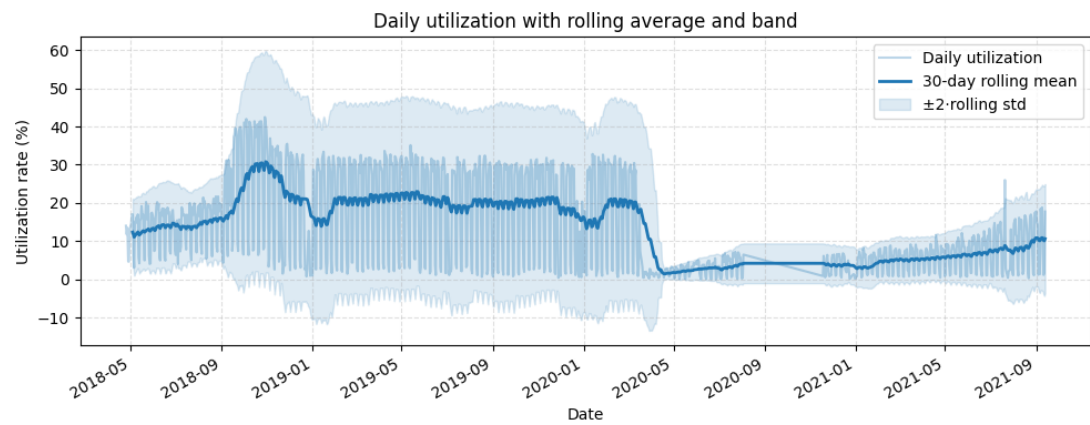


Figure 4: Daily utilization over observation window with rolling average and band

A.3.2 KPI 2: Energy delivered per hour

$$E_T = \sum_s^n \left(E_s \cdot \frac{t_T}{t_s} \right) \quad (2)$$

T = Timeframe in observation window

E_T = Energy delivered in T

E_s = Energy delivered in s

s = Session s

n = Total number of sessions

t_s = Total charging duration of the session (doneChargingTime - connectionTime)

t_T = Charging duration of session in T

Key assumption: Energy is delivered uniformly over the charging duration

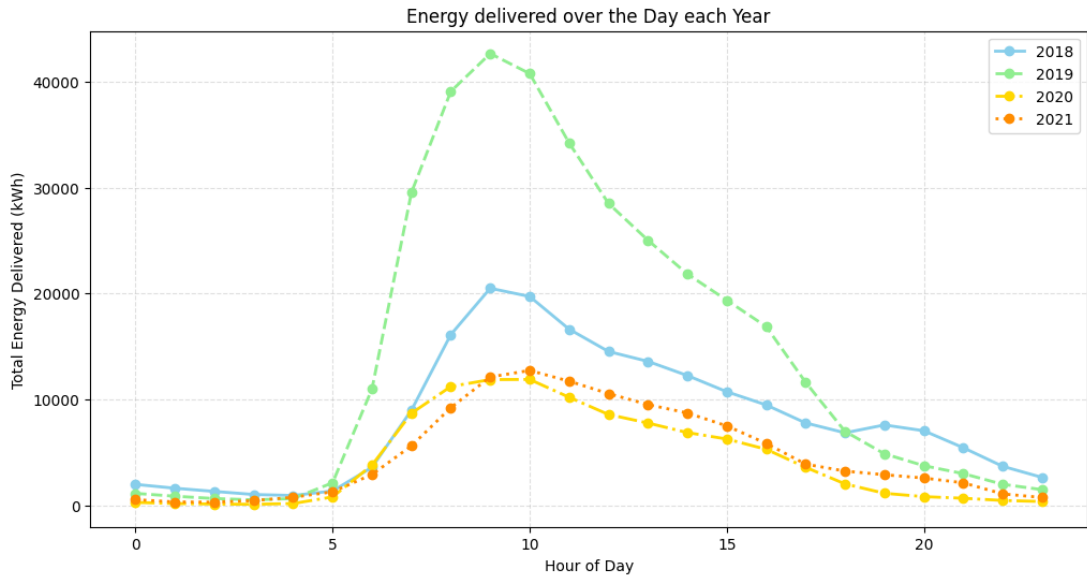


Figure 5: Yearly comparison of total energy delivered by hour of day

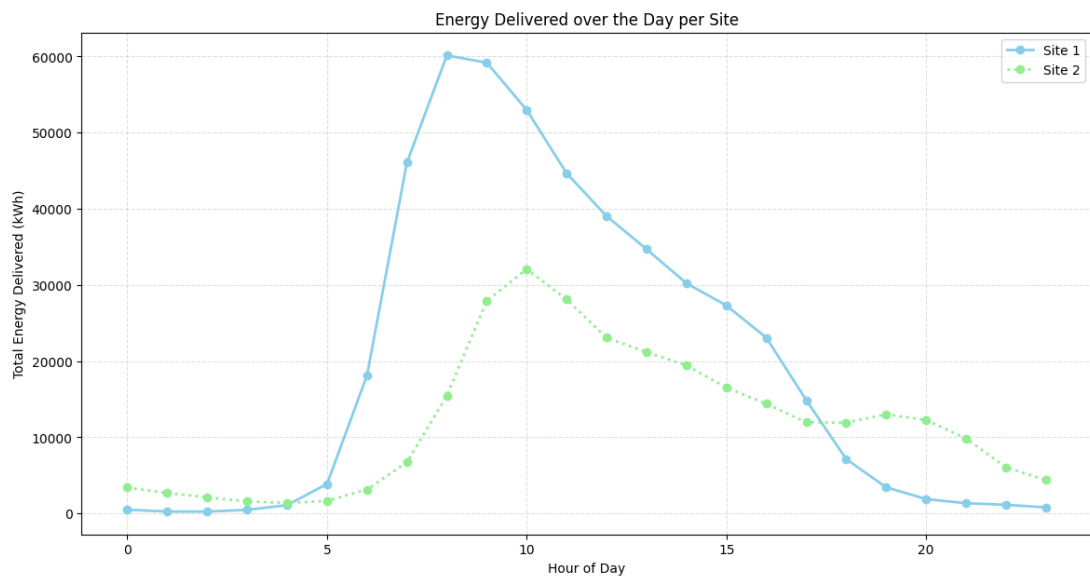


Figure 6: Site comparison of total energy delivered by hour of day

A.3.3 KPI 3: Registration rate

$$R = \frac{S_R}{S} \quad (3)$$

R = Registration Rate

S_R = Number of sessions by registered users

S = Total number of sessions

Portion of sessions with registered vs non-registered users

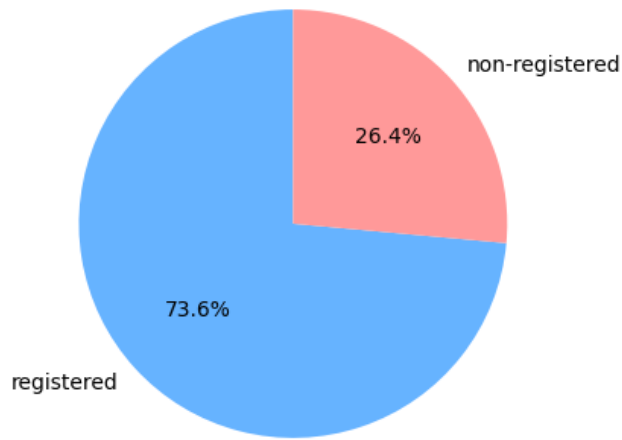


Figure 7: Portion of registered and non-registered user sessions

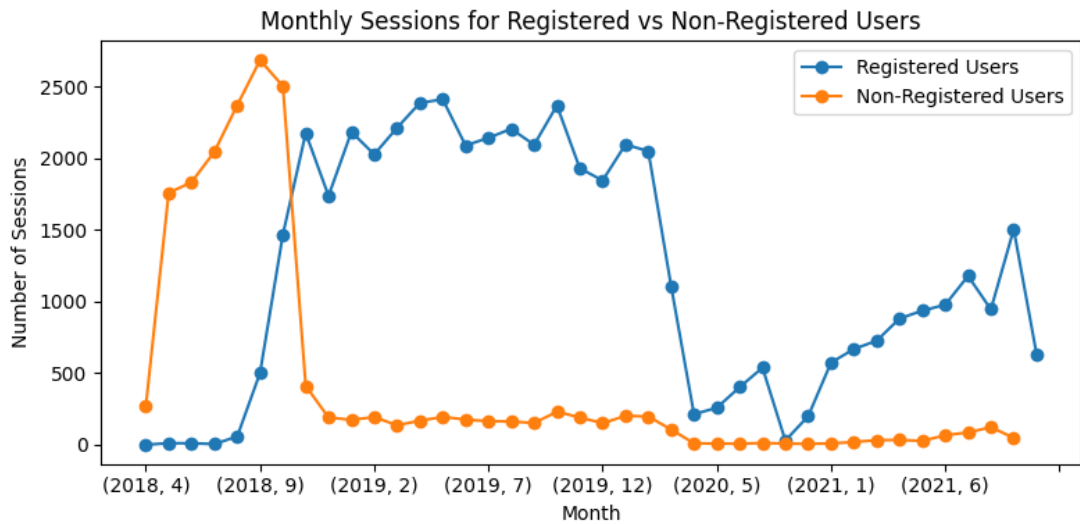


Figure 8: Monthly values for registered vs. non-registered user sessions

A.3.4 Site Characteristics

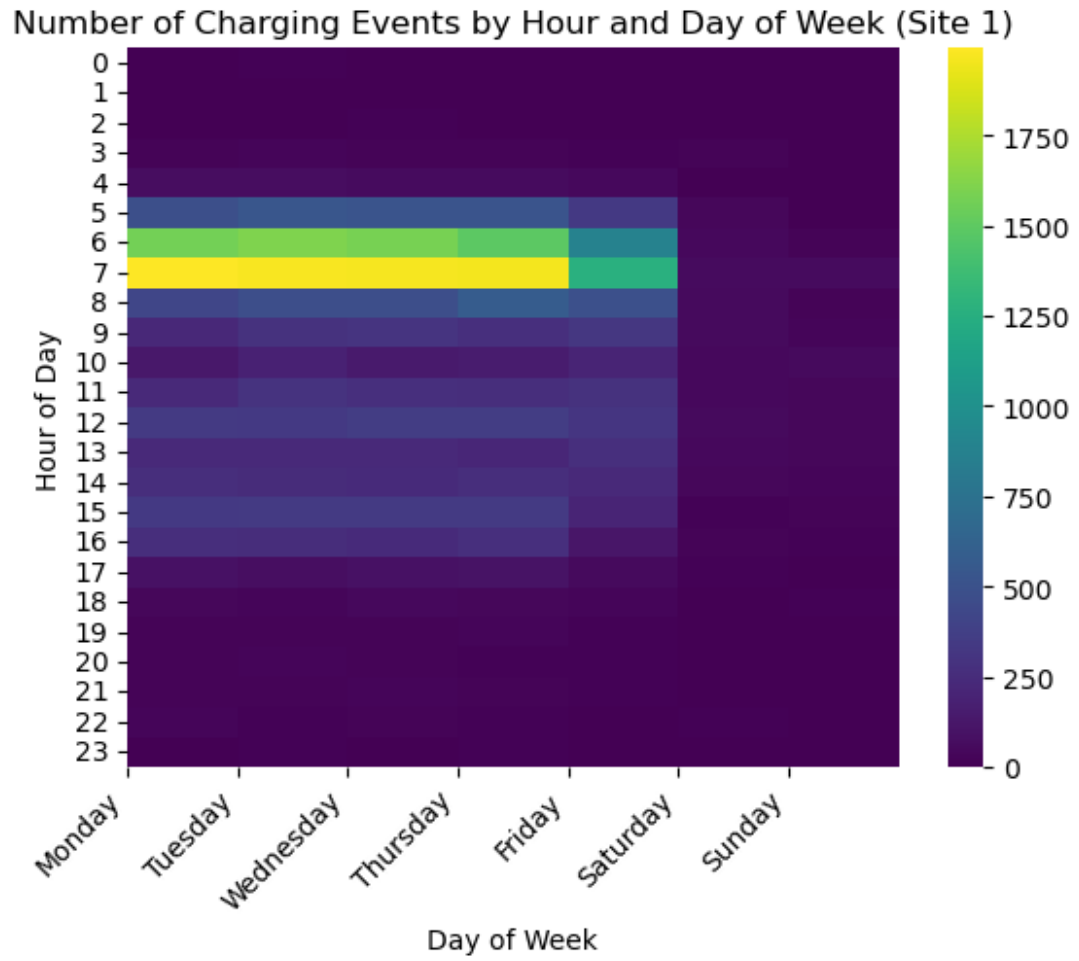


Figure 9: Heatmap for number of charging events by hour and day of week (Site 1)

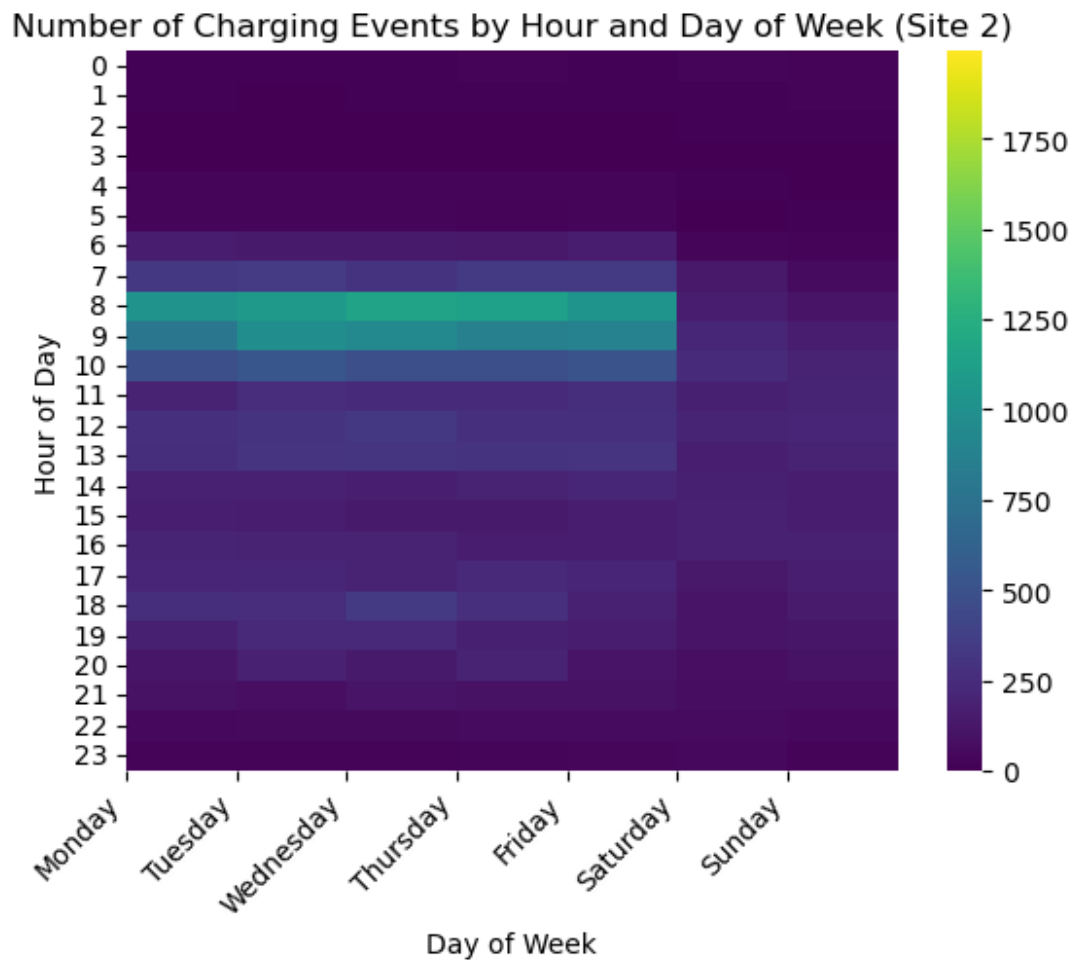


Figure 10: Heatmap for number of charging events by hour and day of week (Site 2)

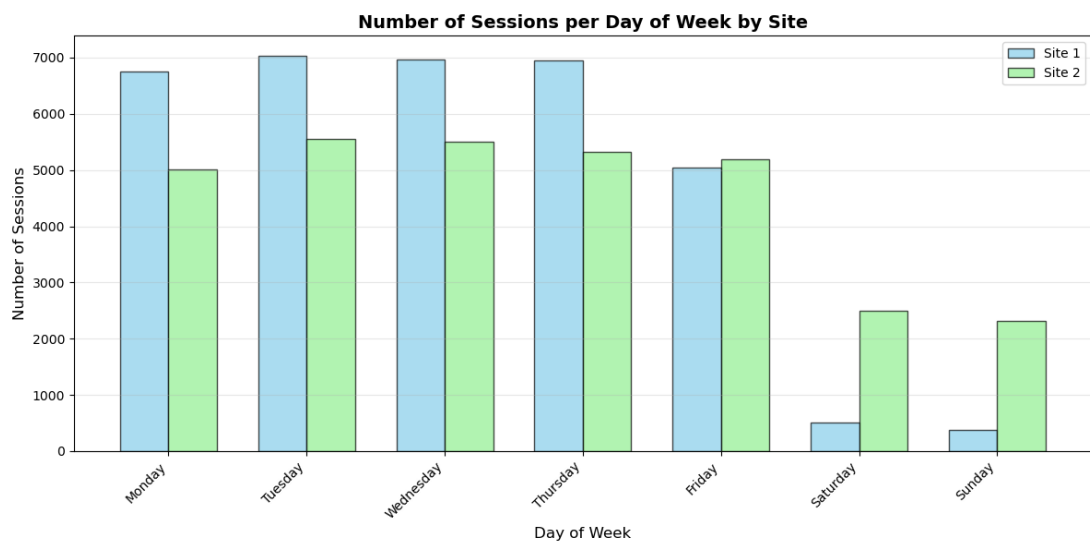


Figure 11: Number of sessions per site by day of week

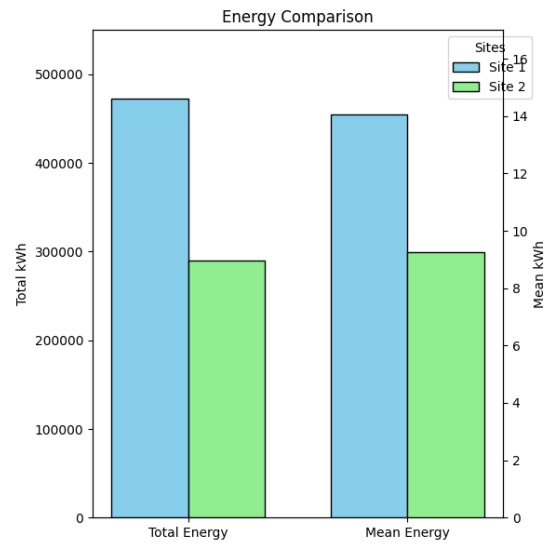


Figure 12: Total energy and mean energy per session delivered per site

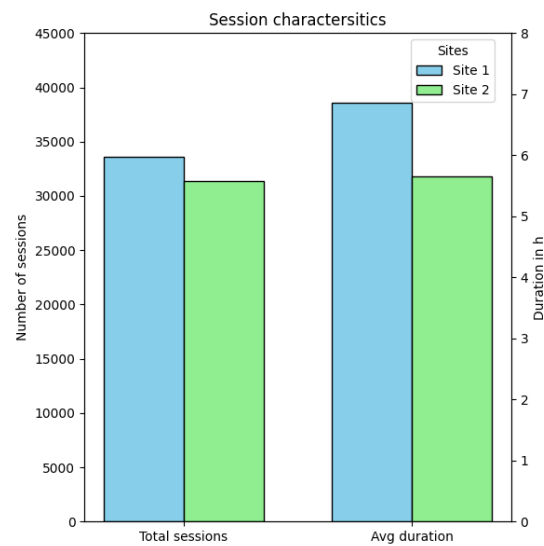


Figure 13: Total sessions and average session duration per site

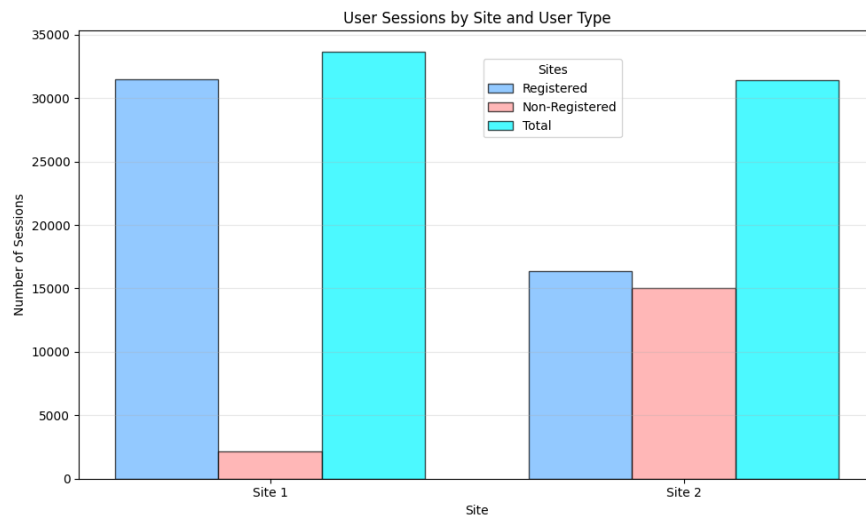


Figure 14: Non-registered vs. registered user share by site

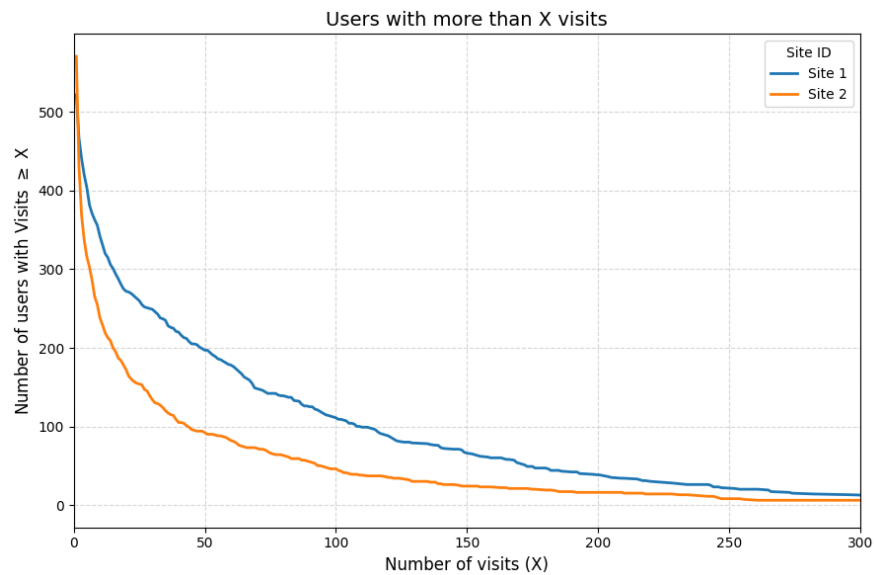


Figure 15: Survival plot for the number of returning users per site

A.3.5 Clustering

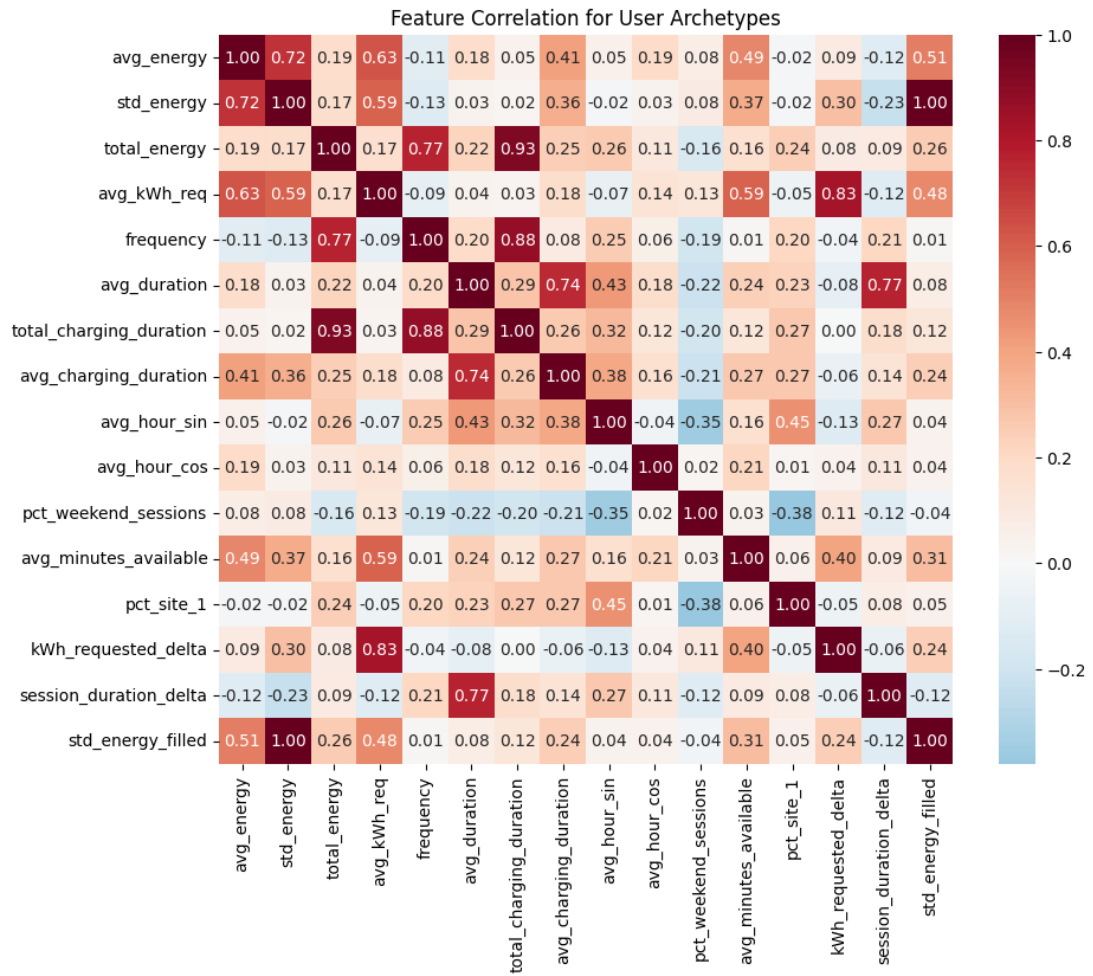


Figure 16: Correlation Matrix used to identify redundant features

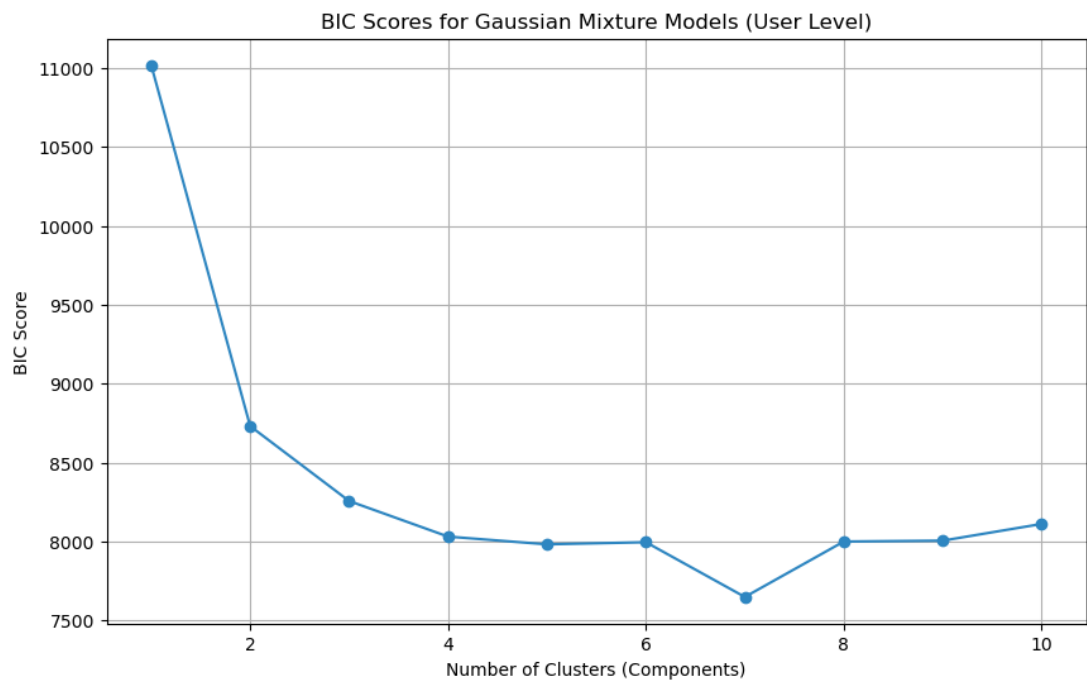


Figure 17: BIC Scores for Gaussian Mixture Models (User Level)

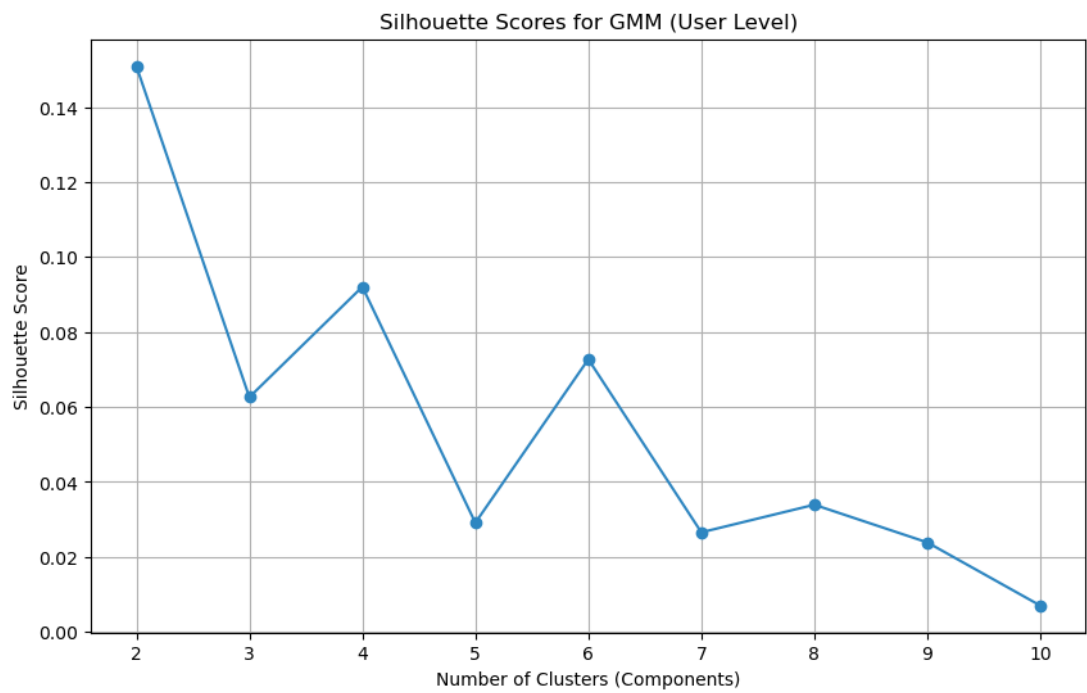


Figure 18: Silhouette Scores for Gaussian Mixture Models (User Level)

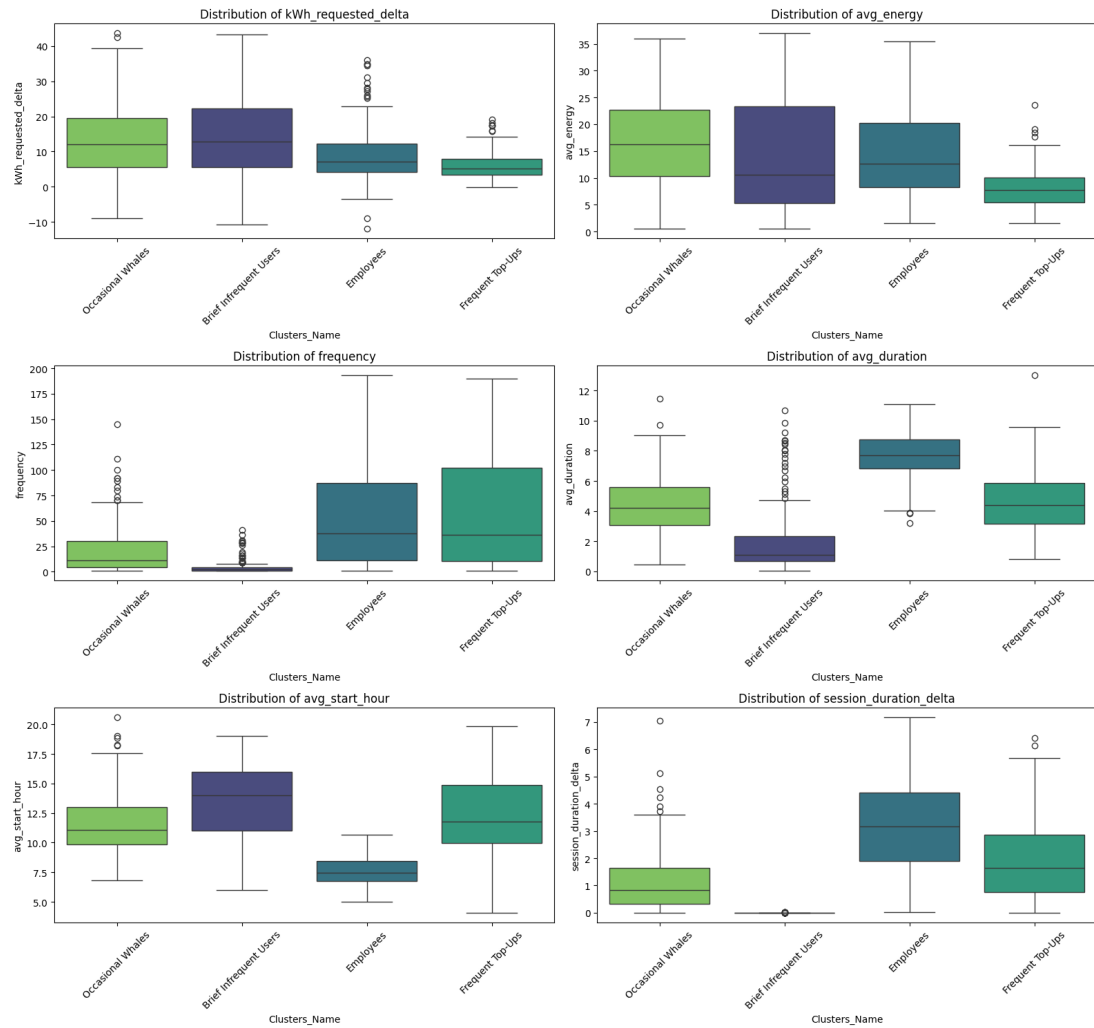


Figure 19: Distribution of feature values in resulting clusters

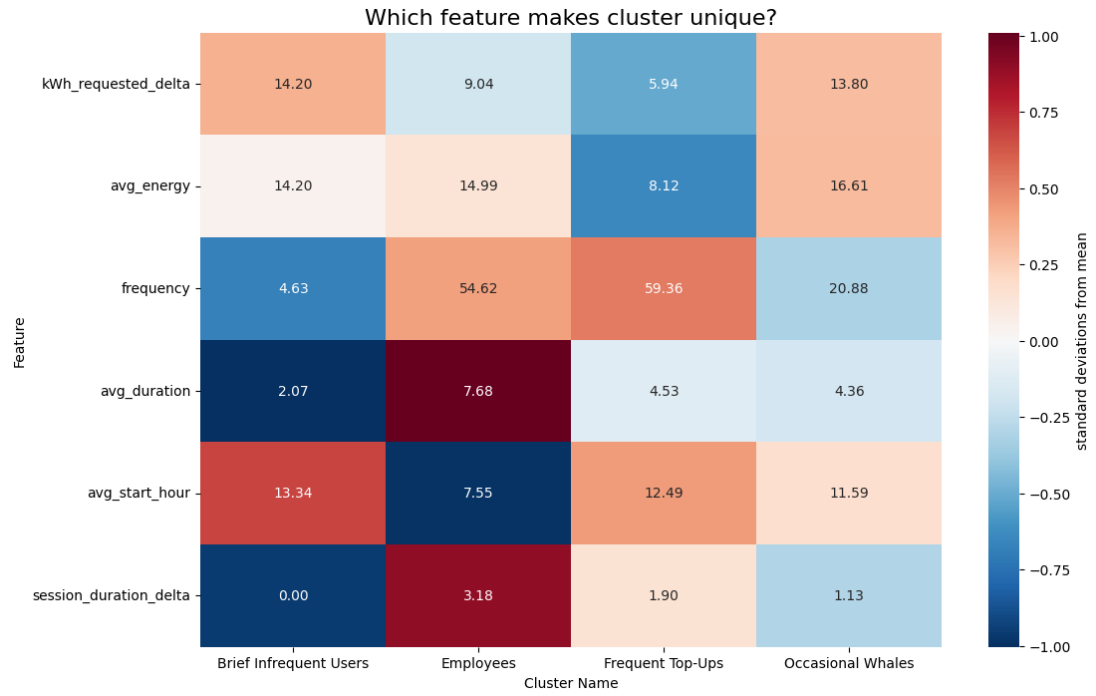


Figure 20: Average values for features in resulting clusters

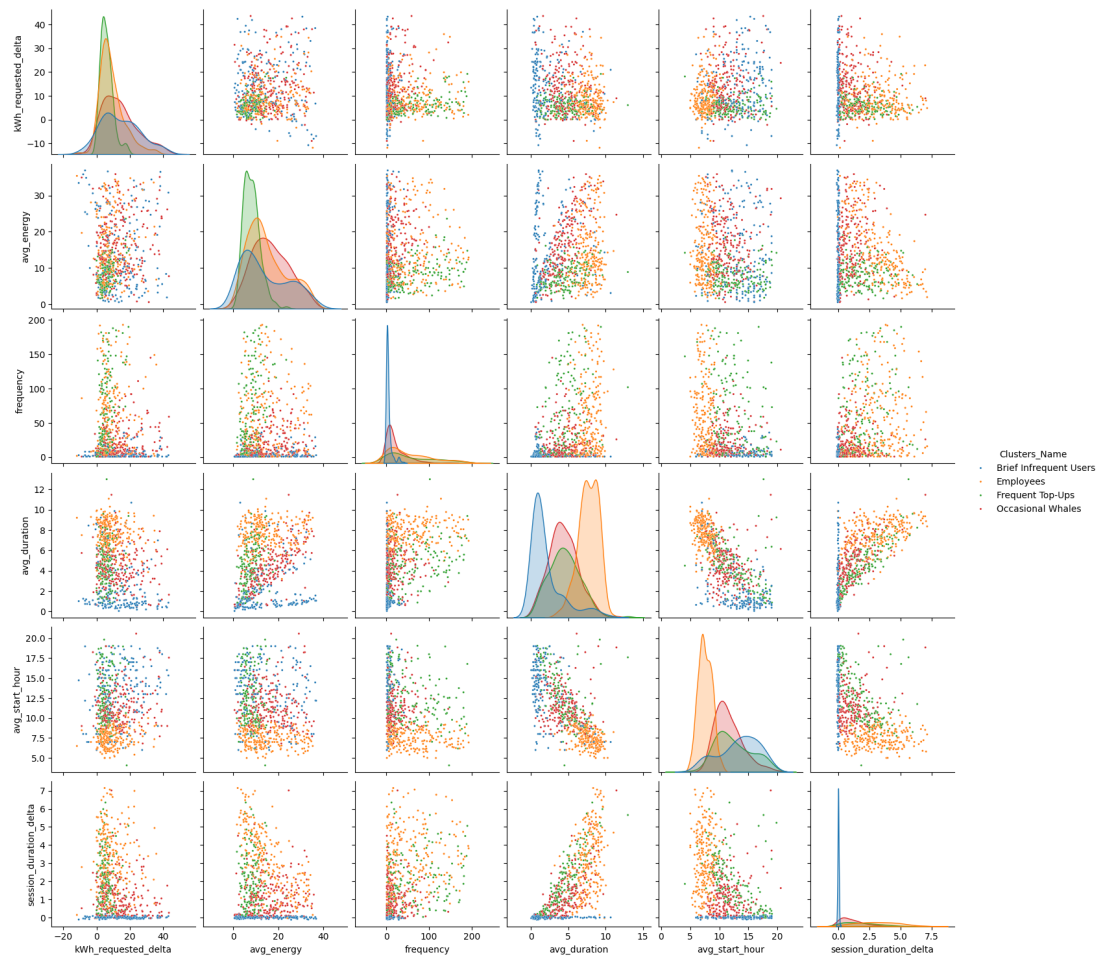


Figure 21: Pairplot of clusters over all features

A.3.6 Utilization Prediction

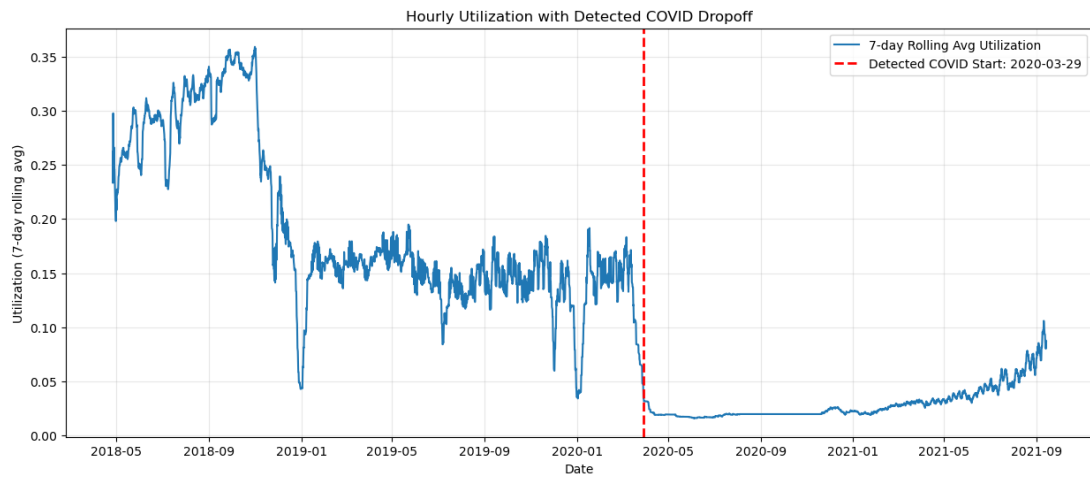


Figure 22: Identification of the start of the covid impact

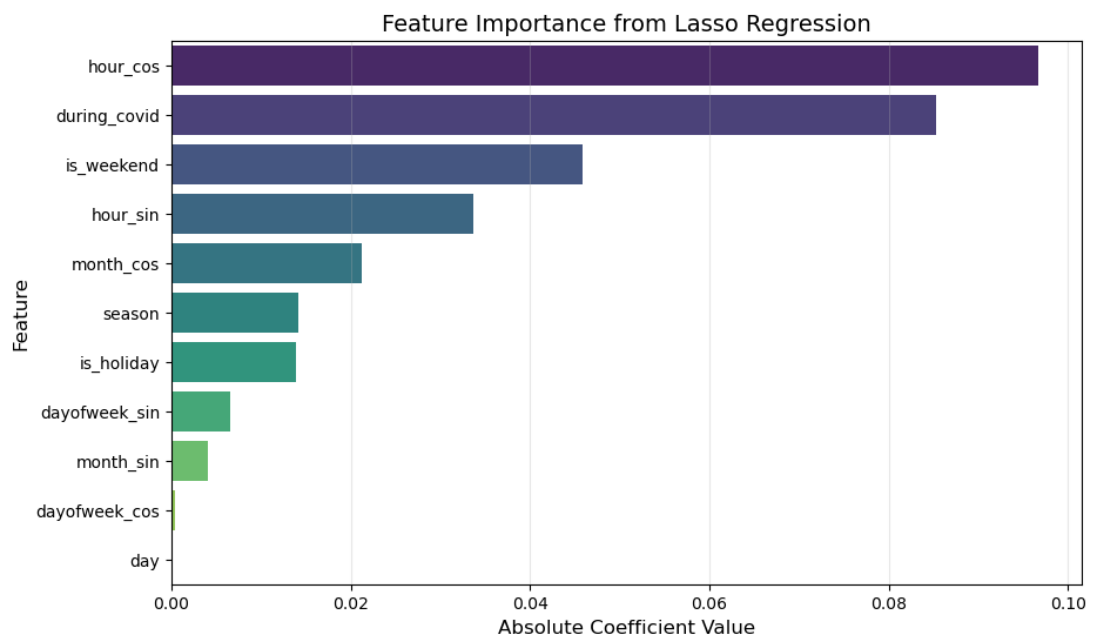


Figure 23: Feature Importance computed by Lasso Feature Selection

Id	Dimension	dtype
0	during_covid	int64
1	is_holiday	int64
2	is_weekend	int64
3	season	int64
4	hour_sin	float64
5	hour_cos	float64
6	dayofweek_sin	float64
7	dayofweek_cos	float64
8	month_sin	float64
9	month_cos	float64

Table 2: Resulting feature set used in model training

Model	Ridge
Degree	5
Alpha	10.0
R2	0.806611
MSE	0.00739

Table 3: Resulting hyperparameters and model type after training and evaluation

Model	R2	RMSE	MAE
Regression Model	0.802238	0.085671	0.058214
Decision Tree	0.712071	0.103373	0.067764
Neural Network	0.784060	0.089522	0.062154

Table 4: Comparison of test metrics for the three models

Metric	Value
Test RMSE	0.0856711772022233
Test R2	0.8022380388502471
Difference in R2 (Holdout vs. Test)	0.44%

Table 5: Ridge regression model performance on test set

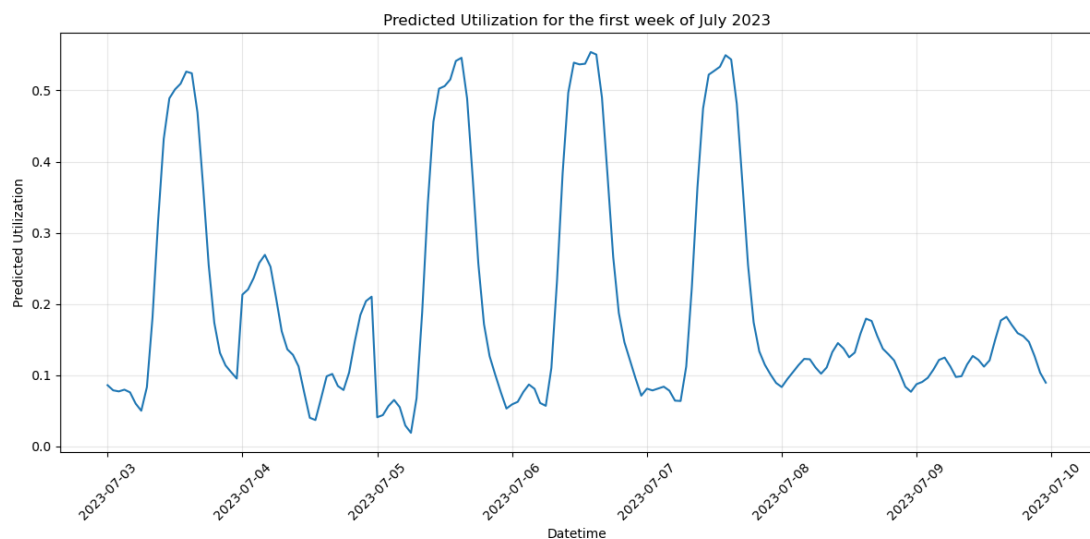


Figure 24: Predicted hourly utilization rate - first week of July 2023

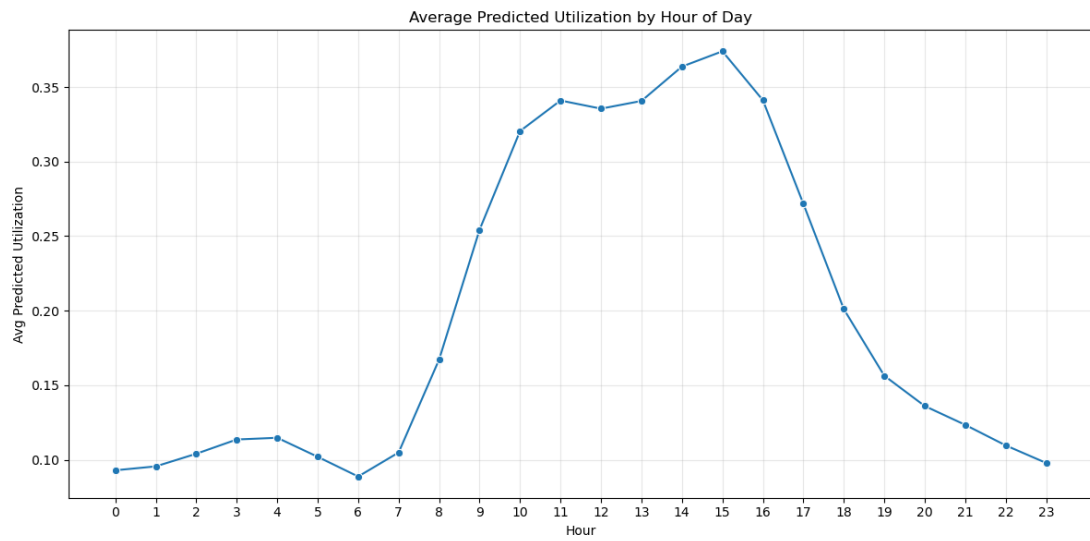


Figure 25: Predicted average utilization rate by hour of day

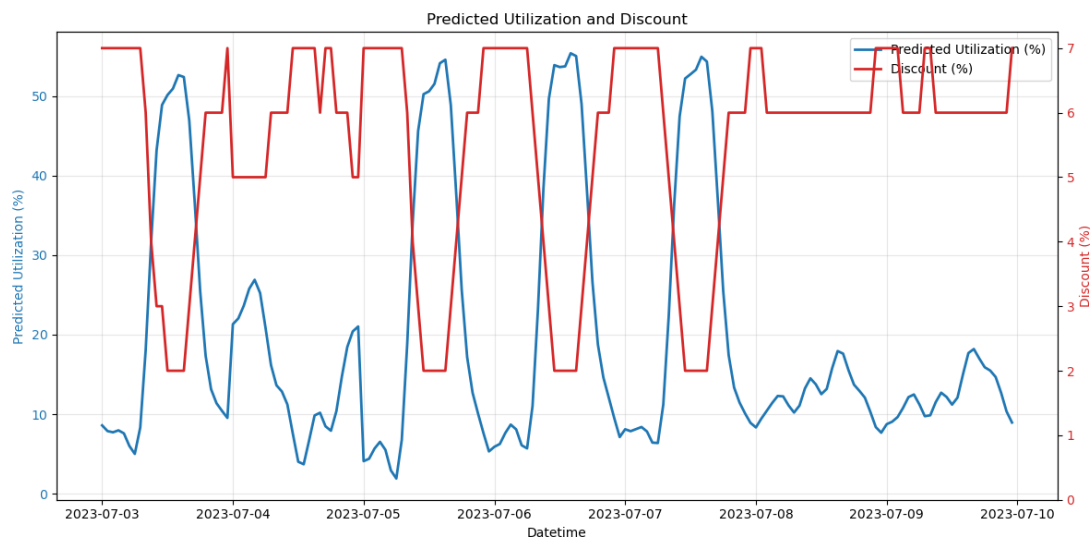


Figure 26: Predicted utilization rate and discount rates - first week of July 2023

A.3.7 Decision Tree

For the Decision Tree Model, we used a Train-Hold-Test-Split (50%/20%/30%). We needed to use a new split because decision trees have automatic feature selection and do not work with cyclic features. First we used gridsearch to find the tree with the best performance and then computed the smallest tree looking for a depth, that does not reduce performance more than 10% compared to the best performing decision tree.

A.3.8 Neural Network

We used the same split as for the regression model. The neural network uses a two-hidden-layer architecture ($64 \rightarrow 32$ neurons) with ReLU activations and a linear output neuron for continuous regression. We compile with the Adam optimizer and MSE loss, and apply early stopping (patience=5) on validation loss as implicit regularization, making dropout unnecessary for this dataset size. EarlyStopping automatically stops the training when the model stops improving.